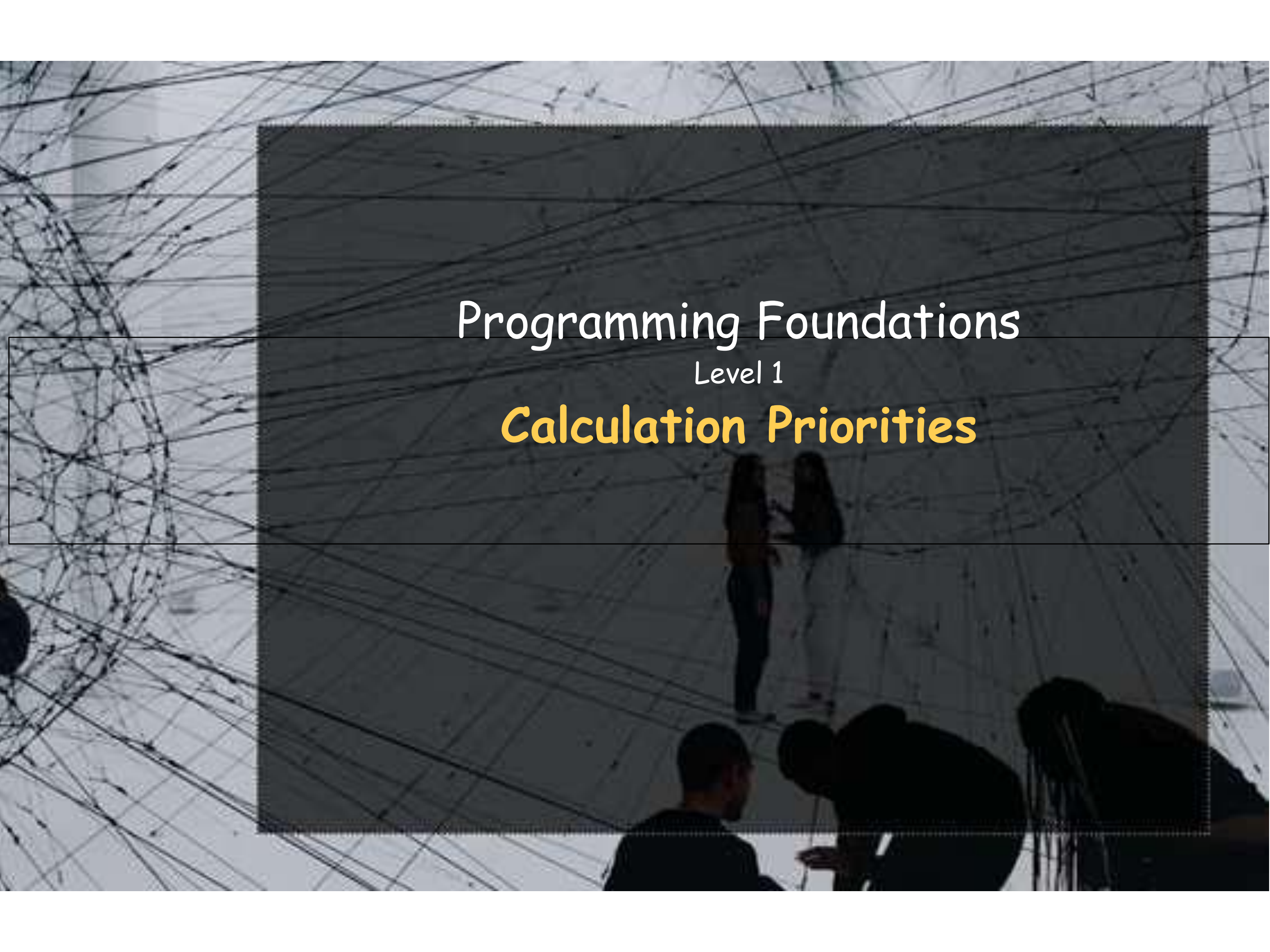




Programming Foundations Level 1



Programming Foundations

Level 1

Calculation Priorities

Calculation Priorities (Math & Logic) — Do It In the Right Order

A huge part of programming (and math!) is **doing operations in the correct order**. This lesson shows how to evaluate expressions correctly using **operation priorities** (parentheses, powers, multiply/divide, add/subtract) and **logic priorities** (NOT, AND, OR) — with step-by-step examples from the slides.

73% of people fail to solve this!

$$4 \times 4 + 4 \times 4 + 4 - 4 \times 4$$

Stop the video now, try to solve it and come back

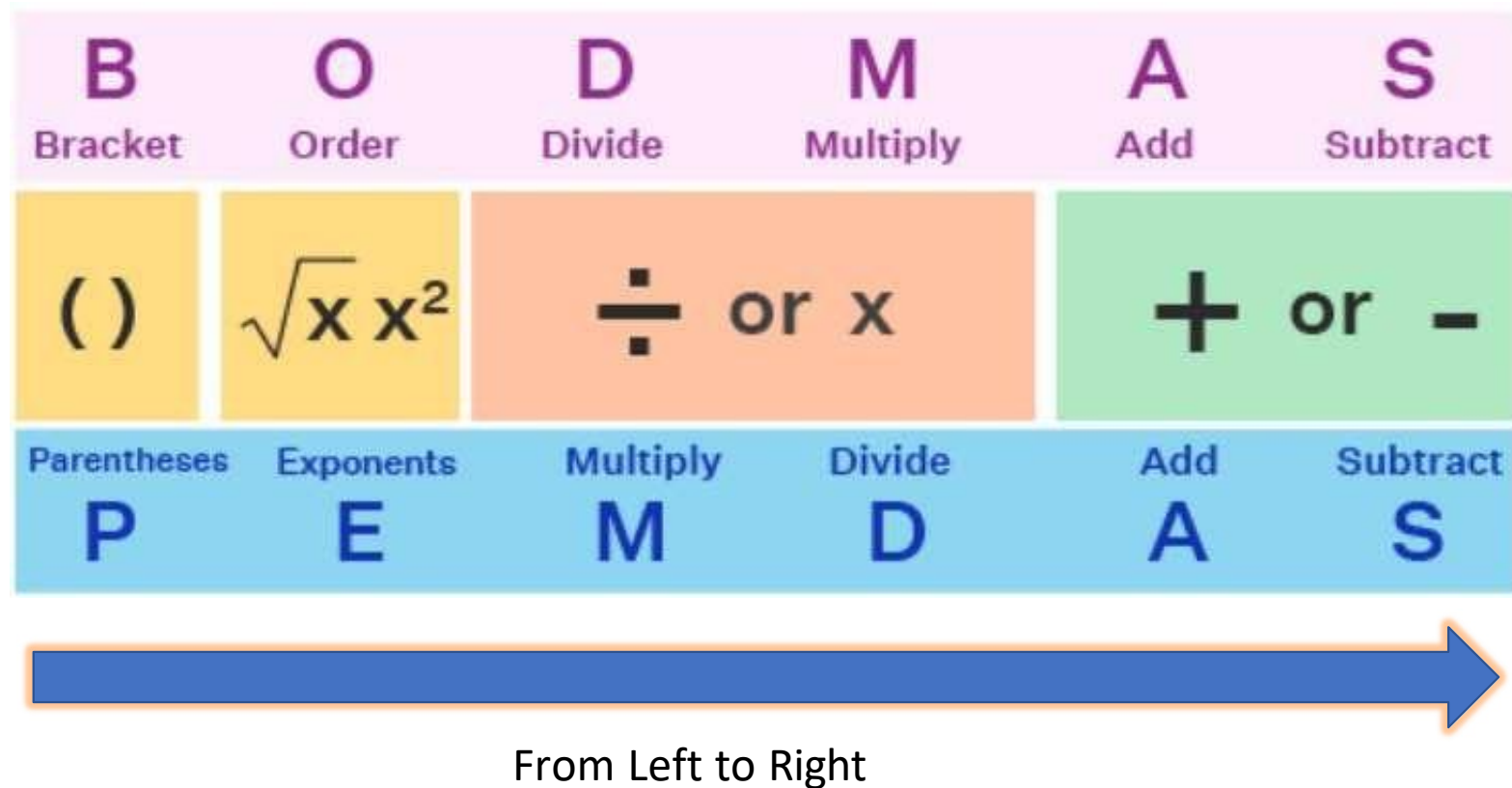
Why?!

The answer is
20

— yet the correct answer is 20 when you respect the rules. (not 36, 52, etc.).

Reason: priority for $\times$ and $\div$ before $+$ and $-$, and for the same rank, go left to right.

Order of Operations



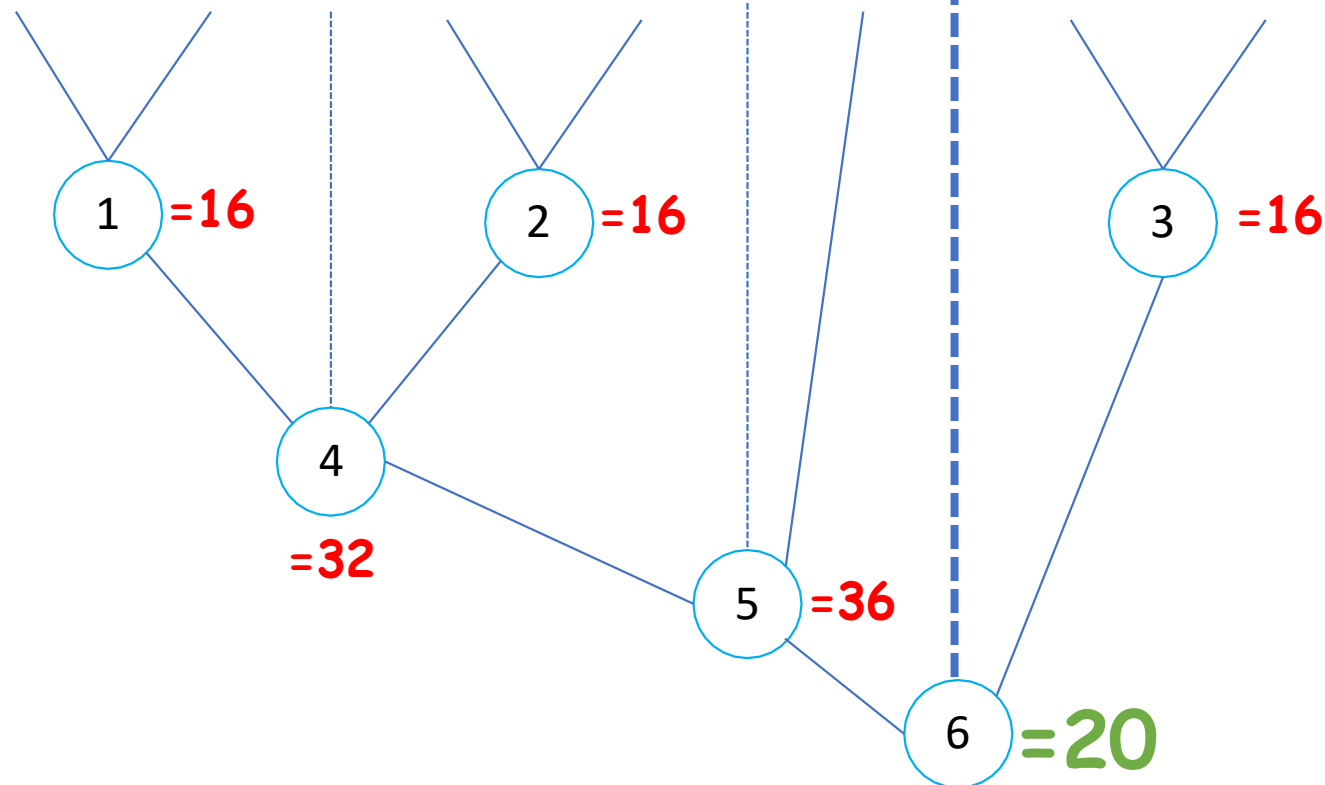
- 1 Parentheses/**B**rackets first
- 2 Exponents/**O**rders (e.g., 2^2)
- 3 **M/D** Multiply & Divide (left→right)
- 4 **A/S** Add & Subtract (left→right)

Keep in mind **parentheses first** and **left-to-right** for same-priority ops.

Example 1:
Solve this:

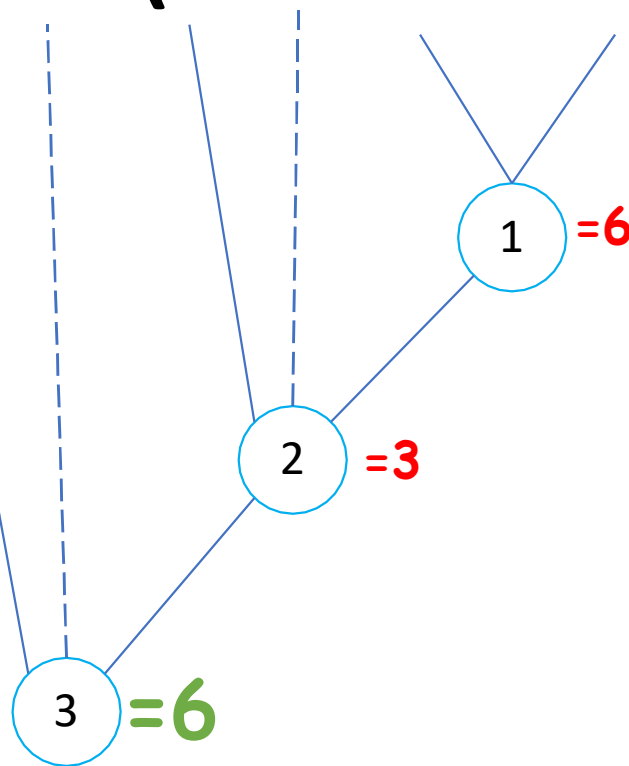
Order of Operations					
B Bracket	O Order	D Divide	M Multiply	A Add	S Subtract
()	$\sqrt{x} x^2$	$\div$ or $\times$		$+$ or $-$	
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S
From Left to Right					

$$4 \times 4 + 4 \times 4 + 4 - 4 \times 4$$



Example 2: Solve this:

$$18 \div (9 - 2 \times 3)$$



Inside parentheses + precedence:

$$18 \div (9 - 2 \times 3)$$

Compute inside: $2 \times 3 = 6 \rightarrow (9 - 6) = 3$

Then: $18 \div 3 = 6$ ✓ (answer = 6).

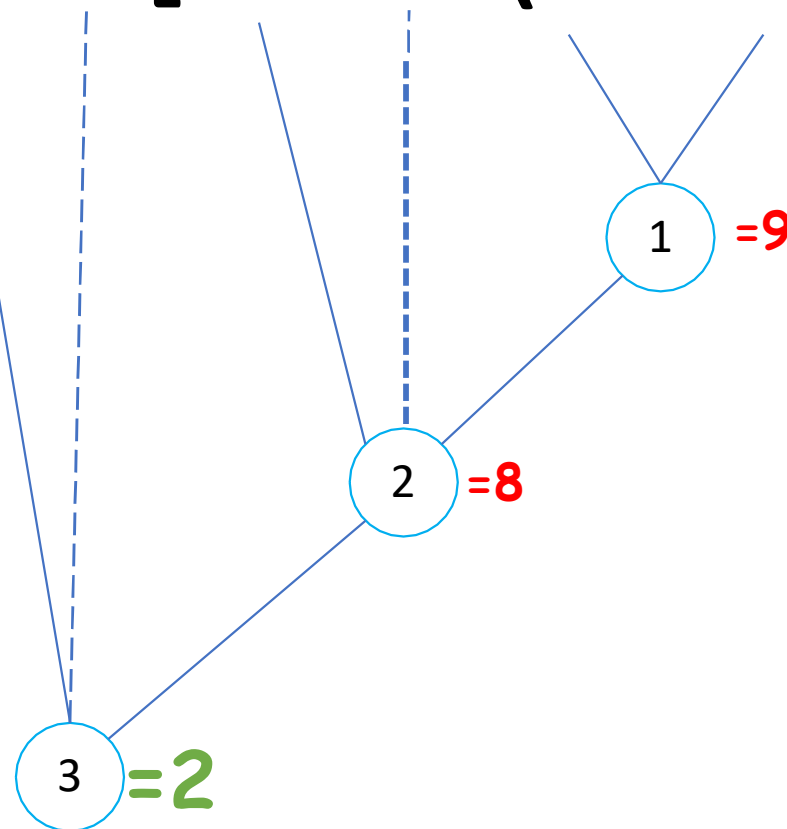
Order of Operations					
B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract
()	$\sqrt{x} x^2$	$\div$ or $\times$		$+$ or $-$	
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S
From Left to Right					

Example 3: Solve this:

Order of Operations					
B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract
()	$\sqrt{x} x^2$	$\div$ or $\times$	$+$ or $-$		
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S

From Left to Right

$$10 - [72 \div (4 + 5)]$$



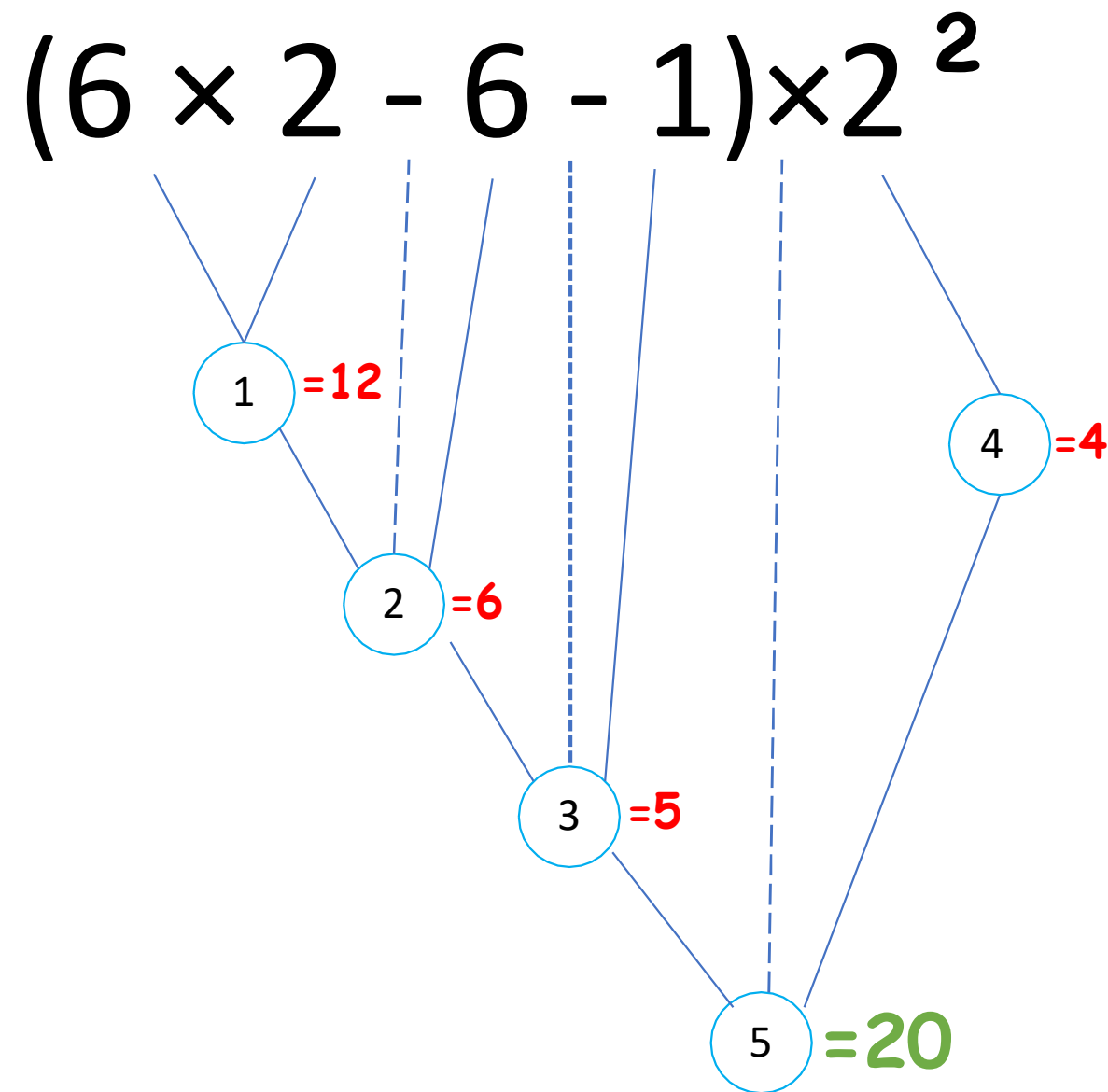
Brackets within brackets:

$$10 - [72 \div (4 + 5)]$$

$(4+5)=9$, so $72 \div 9 = 8$, then $10 - 8 = 2$ ☒ (answer = 2).

Example 4: Solve this:

Order of Operations					
B Bracket	O Order	D Divide	M Multiply	A Add	S Subtract
()	$\sqrt{}$ x^2	$\div$ or $\times$		$+$ or $-$	
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S
From Left to Right					



Powers & grouping:

$$(6 \times 2 - 6 - 1) \times 2^2$$

Inside: $6 \times 2 = 12$, $12 - 6 = 6$, $6 - 1 = 5 \rightarrow$ so $(...) = 5$

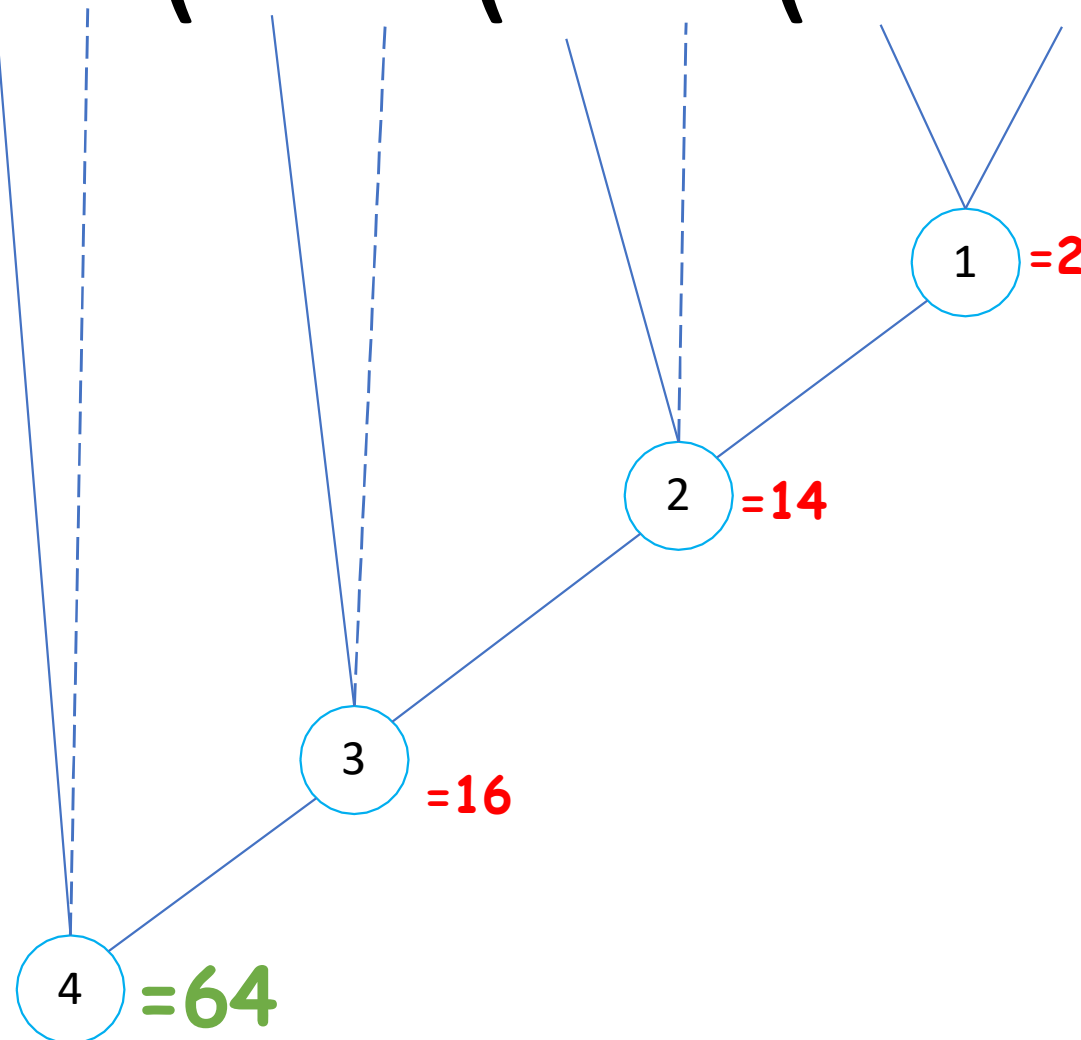
Power: $2^2 = 4 \rightarrow 5 \times 4 = 20$ ✓ (answer = 20).

Example 5: Solve this:

Order of Operations					
B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract
()	$\sqrt{x} x^2$	$\div$ or x	$+$ or $-$		
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S

From Left to Right

$$4 \times (2 + (7 \times (5 - 3)))$$



Nested parentheses:

$$4 \times (2 + (7 \times (5 - 3)))$$

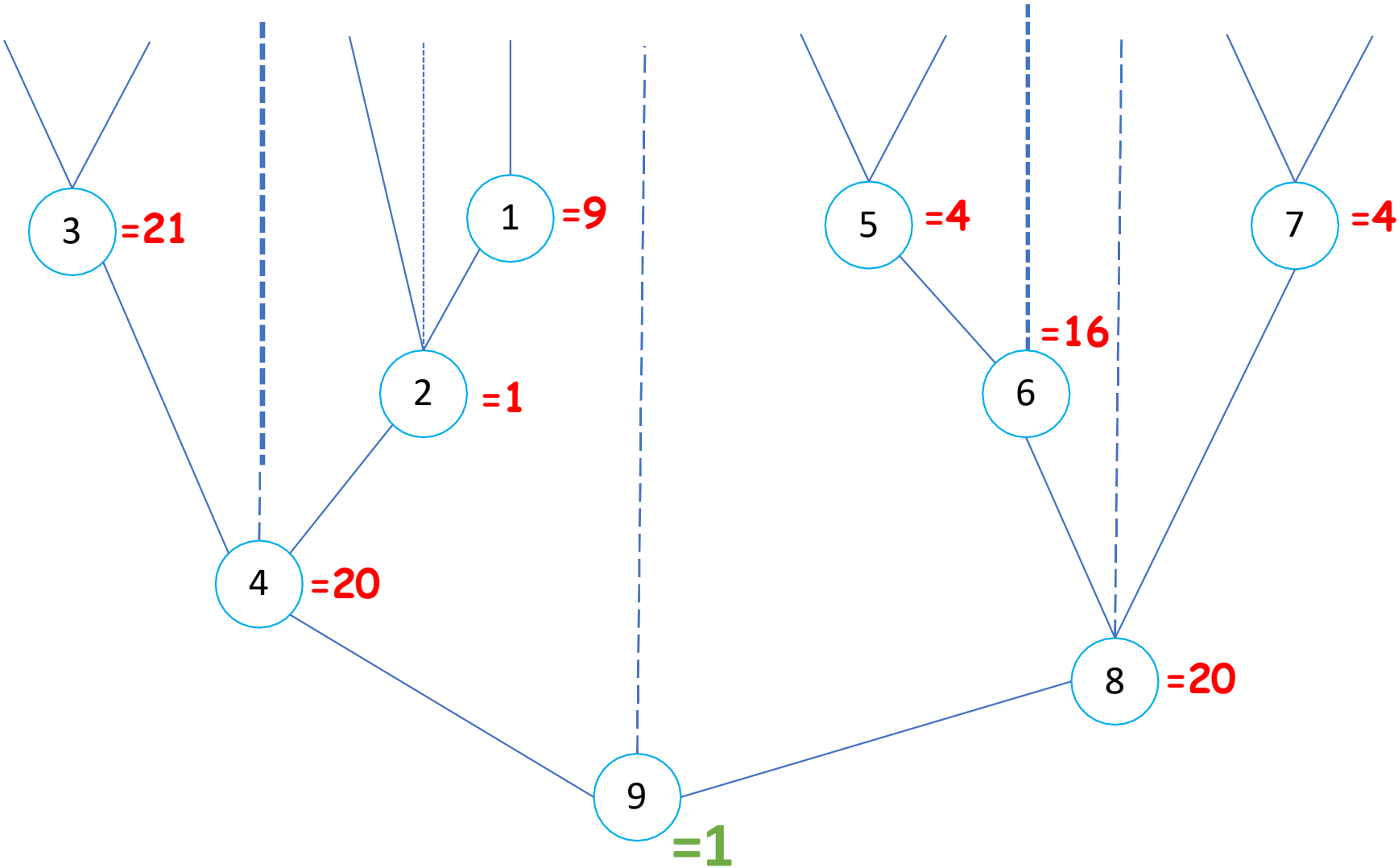
Innermost: $5 - 3 = 2 \rightarrow 7 \times 2 = 14 \rightarrow 2 + 14 = 16 \rightarrow 4 \times 16 = 64$ ✓ (answer = 64).

Example 6:
Solve this:

$$(1 + 20 - 9 \div 3^2) \div ((3 + 1)^2 + 8 \div 2)$$

Order of Operations					
B Bracket	O Order	D Divide	M Multiply	A Add	S Subtract
()	$\sqrt{x} x^2$	$\div$ or $\times$		$+$ or $-$	
Parentheses	Exponents	Multiply	Divide	Add	Subtract
P	E	M	D	A	S

From Left to Right



The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, adding a human element to the technical theme.

Programming Foundations

Level 1

Logic Priorities

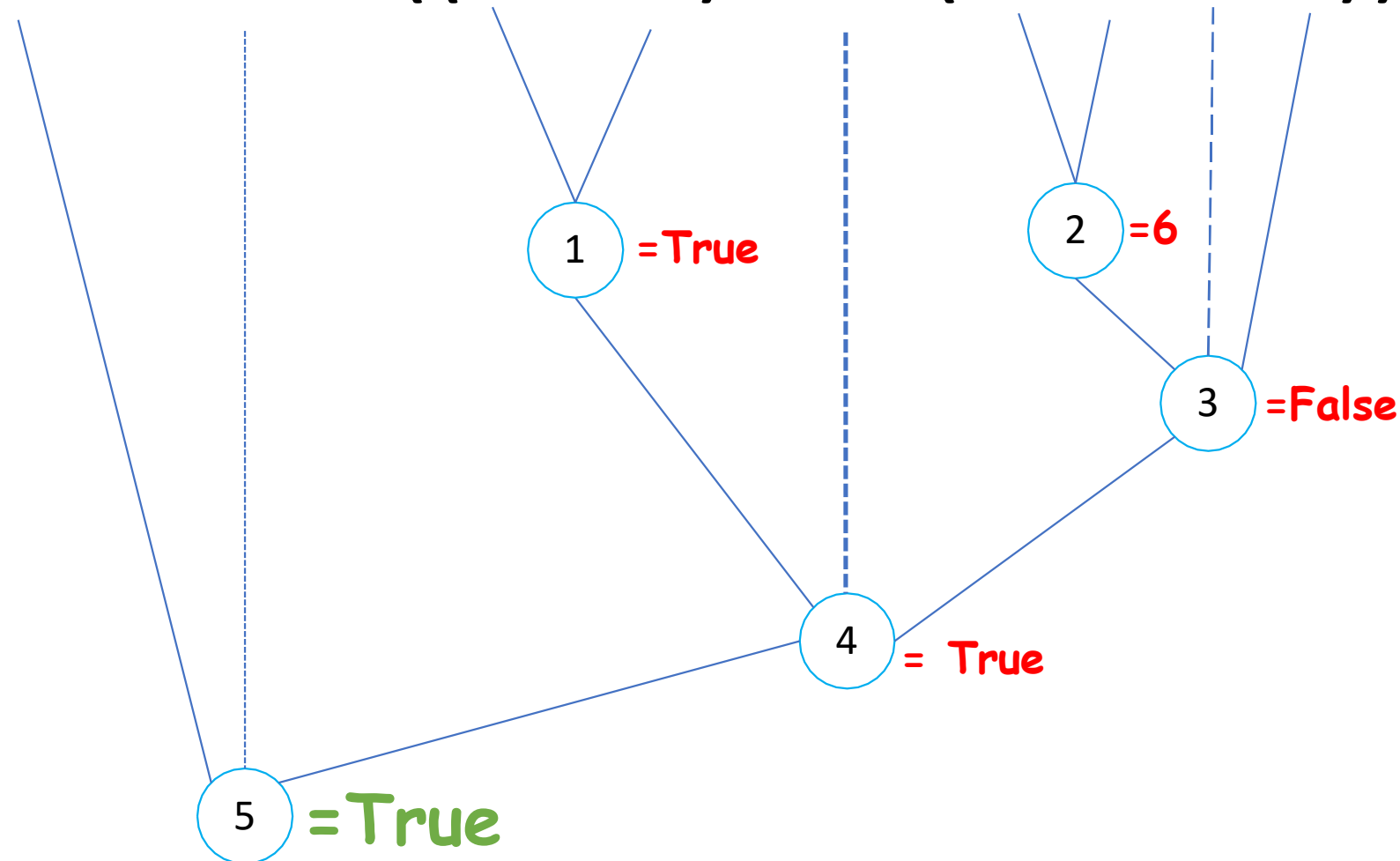
Solve this:

Order of Operations

B	O	D	M	A	S
Bracket	Order	Divide	Multiply	Add	Subtract
()	$\sqrt{x}$ x^2	$\div$ or x	$+$ or $-$		
P	E	M	D	A	S
Parenttheses	Exponents	Multiply	Divide	Add	Subtract

From Left to Right

True AND ((9 > 2) OR (4 + 2 = 5))



True AND ((9 > 2) OR (4 + 2 = 5))

(9 > 2) is **True**; (4 + 2 = 5) is **False** → (True OR False) = **True**

True AND True = **True** ✓ (result = True).



Activity



Instructions:

① Parentheses → ② Powers → ③ Multiply/Divide (left to right) → ④ Add/Subtract (left to right)



Hint:

- 👉 Always start with what's inside parentheses first.
- 👉 Simplify exponents before multiplication/division.
- 👉 Finally, perform addition and subtraction in order from left to right.

◆ Set A

- ① $6 \div 3 + 4 \times 2$
- ② $6 \times 4 - 12 \div 3 - 8$
- ③ $4 \times 4 - 3 \times 3 - 16 \div 4$
- ④ $20 - (3 \times 23 - 5)$
- ⑤ $(5 + 2)^2 - 9 \times 3 + 2^3$
- ⑥ $(7 - \sqrt{9}) \times (42 - 3 + 1)$

◆ Set B

- ① $5 \times (2^2 + 3) + (2^3)^2$
- ② $6 \times 2 + 5 \times 1 + 4 \div 2 - 1$
- ③ $1 + (2 \times (4 - 3 + 1) + 7) - 2$
- ④ $10 + 6 \times (3 - 2) \times 8 \div 24 - 1$
- ⑤ $(20 - 18)^3 \div 8 \times 3 - 1$
- ⑥ $(2 + 1 - 3 + 1 + 4)^2 - 2 \times 4$



Activity Solution



Solution:

◆ Set A :

① $6 \div 3 + 4 \times 2$

$$6 \div 3 = 2 \rightarrow 4 \times 2 = 8 \rightarrow 2 + 8 = 10$$

② $6 \times 4 - 12 \div 3 - 8$

$$6 \times 4 = 24 \rightarrow 12 \div 3 = 4 \rightarrow 24 - 4 - 8 = 12$$

③ $4 \times 4 - 3 \times 3 - 16 \div 4$

$$4 \times 4 = 16 \rightarrow 3 \times 3 = 9 \rightarrow 16 - 9 = 7 \rightarrow 16 \div 4 = 4 \rightarrow 7 - 4 = 3$$

④ $20 - (3 \times 23 - 5)$

$$3 \times 23 = 69 \rightarrow (69 - 5) = 64 \rightarrow 20 - 64 = -44$$

⑤ $(5 + 2)^2 - 9 \times 3 + 2^3$

$$(5 + 2)^2 = 7^2 = 49 \rightarrow 9 \times 3 = 27 \rightarrow 2^3 = 8 \rightarrow 49 - 27 + 8 = 30$$

⑥ $(7 - \sqrt{9}) \times (42 - 3 + 1)$

$$\sqrt{9} = 3 \rightarrow (7 - 3) = 4 \rightarrow (42 - 3 + 1) = 40 \rightarrow 4 \times 40 = 160$$

◆ **Set B :**

① $5 \times (2^2 + 3) + (2^3)^2$

$2^2 = 4 \rightarrow (4 + 3) = 7 \rightarrow 5 \times 7 = 35 \rightarrow 2^3 = 8 \rightarrow 8^2 = 64 \rightarrow 35 + 64 = 99$

② $6 \times 2 + 5 \times 1 + 4 \div 2 - 1$

$6 \times 2 = 12 \rightarrow 5 \times 1 = 5 \rightarrow 4 \div 2 = 2 \rightarrow 12 + 5 + 2 - 1 = 18$

③ $1 + (2 \times (4 - 3 + 1) + 7) - 2$

$(4 - 3 + 1) = 2 \rightarrow 2 \times 2 = 4 \rightarrow 4 + 7 = 11 \rightarrow 1 + 11 - 2 = 10$

④ $10 + 6 \times (3 - 2) \times 8 \div 24 - 1$

$(3 - 2) = 1 \rightarrow 6 \times 1 \times 8 \div 24 = 48 \div 24 = 2 \rightarrow 10 + 2 - 1 = 11$

⑤ $(20 - 18)^3 \div 8 \times 3 - 1$

$(20 - 18) = 2 \rightarrow 2^3 = 8 \rightarrow 8 \div 8 = 1 \rightarrow 1 \times 3 = 3 \rightarrow 3 - 1 = 2$

⑥ $(2 + 1 - 3 + 1 + 4)^2 - 2 \times 4$

$(2 + 1 - 3 + 1 + 4) = 5 \rightarrow 5^2 = 25 \rightarrow 2 \times 4 = 8 \rightarrow 25 - 8 = 17$

Conclusion

- ✓ **Math priorities:** $() \rightarrow \text{powers} \rightarrow \times/\div \rightarrow +/\text{-}$ with **left**→**right** for ties.
- ✓ **Logic priorities:** Parentheses $\rightarrow$ NOT $\rightarrow$ AND $\rightarrow$ OR.
- ✓ Respecting priorities eliminates "trick" errors and aligns your math with how **compilers/interpreters** evaluate expressions.

Thank you 😊